Choice Principles as Graded Where-Clauses

One programming paradigm at every altitude of the Weihrauch lattice

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Abstract

Two companion notes leave a seam between them. One [ChSch] proposes a Curry–Howard correspondence for nondeterminism in which choice principles are the types and fair schedulers are the terms, and lists "the scheduler calculus" as an open problem: the term language does not exist. The other [GWhere] supplies a term-level paradigm — the `where` clause of a for-comprehension, valued not in Booleans but in a resolution algebra $\mathcal{V}$, evaluated per match — and shows that one syntax covers the Boolean, stochastic, quantum and probabilistic-logic regimes. This note joins them. We argue that a graded clause is not a choice function but a *quantifier* in the sense of Escardó–Oliva, that a resolver is a *selection function*, and that the monad morphism between them is exactly the statement that a resolver realises a clause. Attainability of the induced quantifier is then a sharp dividing line, and we compute where it falls. We extend the graded sort with a fixed-point operator — omitted by design in [GWhere] — and show that without it every graded clause sits at the bottom of the hypercomputation stalk. With it, three independent engines of ascent become available: recursion *depth*, candidate *width*, and *reflection*. The first two are incomparable, which is the syntactic shadow of the incomparability of CC and weak König's lemma in the Weihrauch lattice, and hence of that lattice being a lattice and not a chain. We then write down a clause schema for each of LLPO, CC, WKL, WWKL, $\lim$, DC and closed choice on Baire space, each inhabiting the same `where` slot, so that a scheduler's choice type is readable off its clause. Full AC is reachable, by exactly two routes: an *aperture* channel fed by the environment rather than by the term, and *reflection* — and we note that Krivine realises dependent choice with a `quote` instruction which the rho calculus has as a term former. We show that the complex-valued regime is *transverse* to the lattice rather than high in it, that fairness disciplines are clause combinators rather than typing side-conditions, and that the mechanism buying AC is the mechanism that threatens subject reduction — which is the open hinge of [Agcy] in syntactic dress. Every number reported is computed by the accompanying `choicesim.py`.

1 Introduction

1.1 Two notes and the seam between them

The note [ChSch] proposes that the mathematical theory of choice is a declarative language for nondeterminism and the theory of fairness and scheduling its term language: choice principles are the types, schedulers are the terms, running a schedule is normalisation. Its dictionary is suggestive and its metatheory is well-posed — progress is fairness, subject reduction is the statement that scheduling cannot bootstrap an oracle. But its second open problem, P2, is the term language itself: *define the scheduler calculus*. Nothing in the note says what a scheduler looks like as a piece of syntax a programmer writes.

The note [GWhere] was written for a different purpose — to give syntactic expression to the resolution of nondeterminism so that a programmer can arrange quantum interference — and it arrives at a paradigm. The for-comprehension of rholang 1.4 already carries a `where` clause. Let its condition take values not in $\mathbb{B}$ but in a commutative semiring $\mathcal{V}$, and let it be evaluated *per match*, once for each candidate assignment of data to patterns. Then one syntax covers Boolean gating, Gillespie propensities, Born amplitudes, Viterbi decoding, tropical cost-optimisation and PLN-driven inference; the resolver is the choice of $\mathcal{V}$.

The seam is obvious once stated. [GWhere] has a syntax for resolution and no theory of how much power a given resolution costs. [ChSch] has a theory of how much power a resolution costs and no syntax. The essay both descend from [Agcy] says what the join should look like: the choice type $\chi(c)$ of a cut is a semantic invariant, not in general computable, but "approximable from above by *types* … the term assignment of the companion note is the effective approximation of the decoration, which is why a factorisation defined semantically can nonetheless be certified syntactically." A graded clause is exactly such an effective approximation, and this note builds it.

1.2 The thesis

> *One `where` slot suffices at every altitude. What changes as one climbs the choice hierarchy is not the programming construct but the shape of the formula in it: the presence and polarity of a fixed point, the width of the candidate set it ranges over, and whether it quantifies through the namespace or through an aperture. A scheduler's choice type is a syntactic property of its clause.*

We take the Weihrauch lattice [BGP] as the spine rather than the coarse ladder — finite choice, CC, DC, AC — because the interesting formulae live at degrees the coarse ladder does not name — LLPO, WWKL, $\lim$ — and because two of those degrees turn out to be the ones a rho programmer writes by accident.

1.3 Contributions

C1. **The seam, correctly identified.** A graded clause denotes a map $\mathrm{Cand} \to V$: a weighting of a fibre, not a section of it. Clause, choice principle and scheduler stand in the relation *family* / *assertion that a section exists* / *the section* (§2).

C2. **Attainment as the soundness relation.** A clause is a quantifier (K-monad), a resolver is a selection function (J-monad), and the morphism $\varepsilon \mapsto \lambda p.\, p(\varepsilon(p))$ is "this resolver realises this clause". We characterise attainability: it is total exactly when $\oplus$ is idempotent and the winning candidate carries the multiplicative unit, and it fails for every genuinely graded clause over $(+, \times)$, where the induced quantifier is an expectation (§3, Proposition 1, measured).

C3. **Non-attainment is a lift, not a defect.** Where the quantifier is not attained, the resolver selects into a larger codomain: distributions for $\mathbb{R}_{\geq 0}$ (verified by Monte Carlo), amplitudes for $\mathbb{C}$. The $\mathbb{R}_{\geq 0}$ row of the algebra table is the WWKL row of the lattice (§6.5).

C4. **The graded fixed point, and why nothing climbs without it.** The graded sort of [GWhere] is finitary by design. Proposition 2: every fixed-point-free clause has choice type $\perp$. The ascent is the reintroduction of μ and ν (§4).

C5. **Three engines, two of them incomparable.** Depth, width, reflection. Depth and width are Weihrauch-incomparable, which is the syntactic reason the order on scheduler types is a lattice and not a chain (§5).

C6. **The formulae.** A clause schema per degree, all in one `where` slot, with the rho term each sits in (§6, Table 1).

C7. **Full AC, by exactly two routes.** A ceiling theorem for closed terms over a presented store, and the two ways past it: the aperture and reflection. Krivine realises DC with `quote`; the rho calculus has `quote` as a term former (§7).

C8. **Interference is transverse.** A resolver with a non-empty interference deficit is not a choice function at all, so $\mathbb{C}$ is off the lattice rather than high on it (§8).

C9. **Fairness disciplines are clause combinators.** The bootstrap trap of [GWhere] is a fairness violation manufactured by the value algebra, and its repair $\psi \oplus \epsilon$ is weak fairness written as a formula (§9).

C10. **The hinge, sharpened.** The mechanism that buys AC is the mechanism that breaks subject reduction. A stratification judgement recovers it, and AC lives outside the stratified fragment (§10).

1.4 What this note does not do

It does not prove the Weihrauch degrees claimed for the schemas of §6. Each schema is presented with the reduction it is intended to realise and with the resolver obligation it imposes; turning “this clause requires WKL” into “the choice type of this clause is $\equiv_{\mathrm{W}}$ WKL” requires fixing a representation of the candidate family as a point of a represented space, which we set up only informally. It does not present the topos-theoretic setting in which the Diaconescu argument of §7.5 would be a theorem rather than an analogy. And it does not treat the quantale-graded calculus of [FRho], which is the value-algebra point at which the graded fixed point exists by Knaster–Tarski and which therefore deserves its own treatment.

2 The seam

2.1 What a clause denotes

We recall the object from [GWhere]. A *receipt site* r in a term P is a for-comprehension together with the channels its receipts name. A *candidate* at r is an assignment σ of a distinct datum occurrence to each pattern of each group such that every pattern matches its assigned datum; write $\mathrm{Cand}(P, r)$ for the set of candidates and $\mathrm{Out}(r, \sigma)$ for the residual term the firing produces. A *resolution algebra* is a commutative semiring $\mathcal{V} = (V, \oplus, \otimes, 0, 1)$, with conjunction interpreted by $\otimes$ and disjunction by $\oplus$. A graded clause ψ at r denotes

$$[\![\psi]\!]_r : \mathrm{Cand}(P, r) \longrightarrow V.$$

Candidates are indexed by occurrences, not by terms. The classical interpreter sees nothing of the value except whether it vanishes: a communication is enabled exactly when its clause is nonzero, so the crisp language is the $\mathbb{B}$-valued fragment and support adequacy holds by construction.

2.2 What a choice principle asserts

A choice principle asserts that a family of nonempty sets admits a section. In the Weihrauch setting [BGP] the canonical form is *closed choice*: C_X takes a closed subset $A \subseteq X$, given negatively (by an enumeration of basic open sets exhausting its complement), together with the promise $A \neq \emptyset$, and must return a point of A. The degree of C_X depends on X: $C_{\mathbb{N}}$ is countable choice; $C_{2^{\mathbb{N}}} \equiv_{\mathrm{W}}$ WKL; $C_{\mathbb{N}^{\mathbb{N}}}$ is the strongest of the family; $C_2 \equiv_{\mathrm{W}}$ LLPO is the weakest nontrivial one. Reducibility $\leq_{\mathrm{W}}$ orders them and Weihrauch composition $\star$ — solve one problem, then, using its answer, the next — composes them.

2.3 The category error, and the division of labour

It is tempting, and wrong, to try to *write* a choice principle as a clause. A clause weights a fibre. A choice principle asserts a section of one. No map $\mathrm{Cand} \to V$ is a section of Cand, whatever V is.

What a clause can do is *determine the family whose section is at issue*, and therefore certify, syntactically, how hard that section is to produce. That is the division of labour we adopt throughout, and it is the one [Agcy] asks for.

Object	**Role**
graded clause ψ	the family: which candidates, weighted how
choice principle	the assertion that a section exists, uniformly
resolver / scheduler	the section
attainment	the resolver realises the clause
choice type χ	the Weihrauch degree of the assertion

Remark 1 (Why the certificate can be syntactic though the degree is not computable). χ is undecidable at large [Agcy]. But the schemas of §6 bound it from above by inspection: a fixed-point-free clause over a finite candidate set cannot require more than a definable section, whatever else it does. Certification from above is all a type discipline ever offers, and it is what makes "the oracle it imports is written on its face" [ChSch] an implementable slogan.

3 Quantifiers, selections, and attainment

3.1 The two monads

Escardó and Oliva [EO] study two monads on a fixed codomain R. The *quantifier* (continuation) monad

$$K_R X = (X \to R) \to R,$$

and the *selection* monad

$$J_R X = (X \to R) \to X.$$

There is a monad morphism $\overline{(-)} : J_R \to K_R$ given by

$$\overline{\varepsilon}(p) \;=\; p(\varepsilon(p)),$$

and a quantifier ϕ is *attainable* when $\phi = \overline{\varepsilon}$ for some selection function ε. The paradigm example of an attainable quantifier is $\sup$, attained by $\arg\max$; the paradigm example of a non-attainable one is expectation.

The identification we propose is immediate once the two monads are on the table.

Definition 1 (The quantifier of a clause)**.** Let ψ be a graded clause at a receipt site r over $\mathcal{V}$. Its *induced quantifier* is $\phi_\psi : (\mathrm{Cand}(P, r) \to V) \to V$,

$$\phi_\psi(p) \;=\; \bigoplus_{\sigma \in \mathrm{Cand}(P,r)} [\![\psi]\!]_r(\sigma) \otimes p(\sigma).$$

Here p is a *payoff*: an assessment, in $\mathcal{V}$, of what each candidate leads to. The graded modality supplies the canonical payoff, since $[\![\langle K\rangle_{\mathcal{V}}\psi']\!](P) = \sum_{r,\sigma} [\![\psi_r]\!](\sigma) \otimes [\![\psi']\!](\mathrm{Out}(r, \sigma))$ is exactly ϕ_{ψ_r} applied to the payoff $\sigma \mapsto [\![\psi']\!](\mathrm{Out}(r, \sigma))$. So the graded modality of [GWhere] *is* the clause's quantifier eating the value of the future.

Definition 2 (A resolver realises a clause)**.** A *resolver* at r is a selection function $\varepsilon : (\mathrm{Cand}(P, r) \to V) \to \mathrm{Cand}(P, r)$. It *realises* ψ when $\overline{\varepsilon} = \phi_\psi$, that is, when for every payoff p,

$$p(\varepsilon(p)) \;=\; \bigoplus_{\sigma} [\![\psi]\!]_r(\sigma) \otimes p(\sigma).$$

This is the soundness relation the seam needed. It says: the value the clause assigns to the site as a whole is the value actually obtained by the candidate the resolver picks. A resolver that realises its clause is not merely consistent with it; it *achieves* it.

3.2 When is a clause realisable at all?

Proposition 1 (Attainability)**.** *Let ψ be a clause over $\mathcal{V}$ with finite candidate set.*

(i) If $\oplus$ is idempotent and ψ is crisp — that is, $[\![\psi]\!](\sigma) \in \{0, 1\}$ for every σ — then ϕ_ψ is attained by a definable selection function, namely the $\oplus$-optimal candidate of the support.

(ii) If $\oplus$ is idempotent and ψ is graded, then ϕ_ψ is attained at a payoff p if and only if the $\oplus$-winning candidate σ^* carries the multiplicative unit, $[\![\psi]\!](\sigma^*) = 1$.

(iii) If $\oplus$ is not idempotent — in particular for $\mathcal{V} = \mathbb{R}_{\geq 0}$ or $\mathbb{C}$ — then ϕ_ψ is an expectation and is not attained by any selection function into Cand*, except in the degenerate case of a singleton support.*

Proof sketch. (i) With $\oplus$ idempotent, $\phi_\psi(p) = \bigoplus_{\sigma \in \mathrm{supp}(\psi)} p(\sigma)$, which for $\oplus = \max, \min$ or $\vee$ is a value of p on the support; take ε to return the candidate achieving it. (ii) $\phi_\psi(p) = \bigoplus_\sigma [\![\psi]\!](\sigma) \otimes p(\sigma)$. If the winner carries 1 this equals $p(\sigma^*)$; if it carries $v \neq 1$ the value is $v \otimes p(\sigma^*)$, which is not in general in the image of p. (iii) $\sum_\sigma [\![\psi]\!](\sigma) p(\sigma)$ is a strict convex-type combination whenever two candidates carry nonzero weight, hence lies strictly between two values of p and is not among them, generically. $\square$

The proposition is sharp enough to be measured, and we measured it. Over $20{,}000$ random payoffs per row (`choicesim.py`, E1):

Algebra and clause	**attained**
$\mathbb{B}$, crisp ψ	100.00%
Viterbi $(\max, \times)$, crisp ψ	100.00%
Viterbi $(\max, \times)$, graded ψ	73.83%
tropical $(\min, +)$, crisp ψ	100.00%
tropical $(\min, +)$, graded ψ	0.00%
$\mathbb{R}_{\geq 0}$, crisp ψ	0.00%
$\mathbb{R}_{\geq 0}$, graded ψ	0.00%

and the intermediate Viterbi figure is not noise: over the same payoffs, "ϕ_ψ is attained" agrees with "the winning candidate carries the unit" on 100.00% of cases, which is clause (ii) verified.

Remark 2 (Restriction versus discount). Clause (ii) has a reading worth keeping. Over an idempotent $\oplus$, a clause that merely *restricts* — rules candidates in and out — is realisable by a pure term; a clause that *discounts* — prefers one admissible candidate to another — is not. Gating is free. Preference costs.

3.3 Non-attainment is a lift

Clause (iii) looks like bad news for the whole enterprise, since $\mathbb{R}_{\geq 0}$ and $\mathbb{C}$ are the two rows anyone cares about. It is not. The standard response to a non-attainable quantifier is to enlarge the codomain of the selection function, and the enlargements are exactly the resolvers already in use.

- $\mathcal{V} = \mathbb{R}_{\geq 0}$. Lift ε to select into $\mathcal{D}(\mathrm{Cand})$, distributions over candidates, taking $p \mapsto$ the normalised clause. Then $\mathbb{E}[p(\varepsilon(p))] = \phi_\psi(p)/\phi_\psi(\mathbf{1})$, so attainment holds *in expectation*. Verified by Monte Carlo over $400{,}000$ draws: measured 0.590826 against target 0.590000, and 0.390489 against 0.390000 (`choicesim.py`, E1).
- $\mathcal{V} = \mathbb{C}$. No lift into distributions works, because the coefficients are not positive. The Born rule is a lift into $\mathcal{D}(\mathrm{Cand})$ *after* a quadratic map, and §8 shows why that is a different kind of object.

Principle 1 (The lift is the oracle)**.** The extra structure a resolver must have, beyond being a function into the candidate set, is exactly the failure of attainment of its clause's quantifier. Where the clause is crisp over an idempotent $\oplus$, the scheduler is a pure term and imports nothing. Where it discounts, the scheduler must optimise. Where $\oplus$ is additive, the scheduler must randomise. This is Principle 2's first payoff: the altitude is legible in the algebra before any fixed point is written.

3.4 The product is the composition

Escardó–Oliva's binary product of selection functions,

$$(\varepsilon_0 \otimes \varepsilon_1)(p) = (a_0, a_1), \quad a_0 = \varepsilon_0\big(\lambda x.\, p(x, \varepsilon_1(\lambda y.\, p(x, y)))\big), \quad a_1 = \varepsilon_1(\lambda y.\, p(a_0, y)),$$

is their characteristic operation, and its infinite iteration is bar recursion. Three things now want to be the same operation: this product; Weihrauch composition $\star$, "solve one, then, using its answer, the next" [BGP], which [ChSch] nominates as the cut of the logic of choice; and the sequential chaining of two graded receipt sites through $\langle K\rangle_{\mathcal{V}}$, in which the second site's clause is evaluated against the term the first left behind.

Conjecture 1 (One operation, three names)**.** *For clauses whose quantifiers are attainable, the Escardó–Oliva product of the realising resolvers, the Weihrauch composition of the choice types, and the chaining of the graded sites through $\langle K\rangle_{\mathcal{V}}$ agree. Consequently* DC *is the ω-fold product, bar recursion is its realiser, and the graded modality is the syntax of both.*

We verified the arithmetic half over Viterbi: the product and the chained sites select the same pair on $20{,}000$ random payoff tables, with 0 disagreements. That agreement is a consistency check rather than a discovery — but it is not vacuous, and we made sure it is not. The identification requires the chaining to aggregate in the *same* algebra the selection function optimises in. Chaining with $\oplus = +$ while still selecting by $\arg\max$ — backward induction over expectation rather than over the maximum — disagrees on $4{,}483$ of the same $20{,}000$ tables (`choicesim.py`, E2). So the identification has content, and the content is that the algebra of the clause and the discipline of the scheduler must match.

4 The graded fixed point

4.1 Why the grammar as it stands cannot climb

The graded sort of [GWhere] is

$$\psi ::= \lceil\beta\rceil \mid v \mid \psi\otimes\psi \mid \psi\oplus\psi \mid \langle K\rangle_{\mathcal{V}}\psi \qquad (v\in V),$$

with the crisp sort β carrying full Boolean structure, the spatial connectives, the modality and fixed points. The graded sort deliberately has neither negation nor a fixed point. The exclusion was principled — $\mathbb{C}$ has no order, so Knaster–Tarski is unavailable and a graded ν is a convergence question rather than a definition — but it has a consequence the note did not draw.

Proposition 2 (Everything finitary sits at the bottom)**.** *Let ψ be a graded clause containing no fixed point, at a receipt site whose candidate set is finite. Then ϕ_ψ is computable from the store, and the resolver realising it (under Proposition 1(i), or its lift under §3.3) imports no oracle: $\chi = \bot$.*

Proof sketch. $\lceil\beta\rceil$ is decidable against a finite term by hypothesis on the crisp fragment; scalars are constants; $\otimes$ and $\oplus$ are pointwise; and $\langle K\rangle_{\mathcal{V}}$ applied a syntactically bounded number of times ranges over a finite set of successors, since each site has finitely many candidates and the nesting depth of the modality in ψ is finite. So $[\![\psi]\!]_r$ is a computable finite vector, and choice from a finite presented family is definable outright. □

Corollary 1. *The whole of [GWhere] — including its Shor construction, whose extractable fragment is characterised there as the modality-free clauses with isometric values — lives at the base of the hypercomputation stalk. The extractable fragment and the constructively-choosable fragment coincide.*

That coincidence is worth pausing on. A clause is compilable to a circuit exactly when its resolution is definable; the reason the ZX pipeline of [cwZX] can consume the fragment it consumes is that the fragment demands no choice. Extraction and constructivity are the same constraint seen from two ends.

4.2 The grammar, extended

Definition 3 (Graded formulae with fixed points)**.** Fix a resolution algebra $\mathcal{V}$ and a set of formula variables X. The graded sort is

$$\psi ::= \lceil\beta\rceil \mid v \mid X \mid \psi \otimes \psi \mid \psi \oplus \psi \mid \langle K\rangle_{\mathcal{V}}\psi \mid \langle @\rangle_{\mathcal{V}}\psi \mid \mu X.\, \psi \mid \nu X.\, \psi,$$

subject to the *guardedness* condition that every free occurrence of X in $\mu X.\psi$ or $\nu X.\psi$ lies under a $\langle K\rangle_{\mathcal{V}}$ or a $\langle @\rangle_{\mathcal{V}}$. The new modality $\langle @\rangle_{\mathcal{V}}\psi$ is the graded namespace connective: evaluated at a candidate binding a name y, it returns the value of ψ at the process $*y$ quoted by that name.

Guardedness is what makes the unfolding a genuine iteration of the transfer operator T of [GWhere], rather than a self-reference with no dynamics. The two polarities have distinct readings.

- $\mu X.\psi$ is a *search*: the least fixed point, the value accumulated by those unfoldings that terminate. Operationally, "keep unfolding until the body contributes something nonzero". A μ whose unfolding depth is not bounded by the term is an unbounded search, and unbounded search is where CC enters.
- $\nu X.\psi$ is a *survival* condition: the greatest fixed point, the value carried by unfoldings that never fail. Operationally, "this holds along some infinite path". A ν over an infinitely deep candidate tree is a statement about branches, and statements about branches of infinite trees are where König's lemma enters.

4.3 When the fixed point exists

This is where the value algebra stops being a free parameter.

Proposition 3 (Existence, by algebra)**.** *Let Ψ be the guarded functional on $\mathcal{V}$-valued assignments determined by ψ.*

(i) If $\mathcal{V}$ carries a complete order compatible with $\oplus, \otimes$ — a quantale, as in [FRho] — then μ and ν exist by Knaster–Tarski, and no budget is required.

(ii) If $\mathcal{V} = \mathbb{R}_{\geq 0}$, Ψ is monotone but the lattice is not complete; $\mu X.\psi$ exists as a supremum in $[0, \infty]$ and $\nu X.\psi$ requires a convergence hypothesis. Sufficient: the spectral radius of the transfer operator restricted to the reachable set is < 1.

(iii) If $\mathcal{V} = \mathbb{C}$ there is no order at all. Neither fixed point is defined by approximation from below or above; both must be given as solutions of a linear system, and existence is a spectral condition on T.

Remark 3 (The budget is the honest default). Rholang 1.4 already reins in infinitary property checks with a token budget [Omni]. The budget is precisely the retreat from (i) to Proposition 2: a budgeted μ or ν is a bounded unfolding, hence finitary, hence at $\perp$. This gives the implementation a dial that is also a *semantic* dial — spend budget, gain altitude — and it means a shard can refuse to run a clause whose declared altitude exceeds its policy simply by capping unfoldings.

Principle 2 (The two gradings are not orthogonal)**.** There are two gradings in play: the strength of the choice principle (a Weihrauch degree, an altitude in the stalk of [Agcy]) and the value algebra of the clause. They are transverse in general — $\mathbb{C}$ is not "stronger choice", it is a different resolver — but the fixed point couples them, because whether a fixed point exists at all is a property of the algebra's order structure (Proposition 3). The altitude a clause certifies is a function of formula shape *and* algebra.

5 Three engines of ascent

With the fixed point available, exactly three things can make a clause's family non-presented. We name them and then show that two of them are incomparable.

Definition 4 (The engines)**.** Let ψ be a graded clause at a site r of a term P.

- *Depth.* The *unfolding rank* of ψ is the least n such that ψ's fixed points stabilise within n applications of the transfer operator, or ω if no such n exists. Depth ω with bounded branching is the *depth engine*.
- *Width.* The *candidate width* of r is $\sup |\mathrm{Cand}(P', r)|$ over terms P' reachable from P. Width ω at bounded depth is the *width engine*.
- *Reflection.* ψ uses the *reflection engine* when it contains $\langle @ \rangle_{\mathcal{V}}$ applied to a name bound by the site's own receipts, so that the family it indexes is determined by processes received at run time rather than by sub-terms of P.

Each engine has a plain operational meaning in rholang. Depth ω is a recursive process — built, since core rho has no primitive replication, by the reflection idiom, a receipt that reinstalls itself by dropping its own quotation. Width ω is a channel on which unboundedly many messages accumulate: a producer under recursion whose consumer is slower, or a persistent producer. Reflection is a clause that talks about what it just received.

Proposition 4 (Depth and width are incomparable)**.** *The depth engine at bounded width realises closed choice on a compact space; the width engine at bounded depth realises closed choice on the naturals. These are Weihrauch-incomparable: $C_{\mathbb{N}} \not\leq_{\mathrm{W}} C_{2^{\mathbb{N}}}$ and $C_{2^{\mathbb{N}}} \not\leq_{\mathrm{W}} C_{\mathbb{N}}$ [BGP].*

Proof sketch. Bounded width k and unbounded depth present the candidate family as an infinite, finitely-branching tree whose nodes are the successive sites' candidates and whose branches are the runs a ν quantifies over. A ν-clause asserts the existence of an infinite branch on which the body never vanishes, and by guardedness the set of such branches is closed in $k^{\mathbb{N}}$; choosing one is $C_{k^{\mathbb{N}}} \equiv_{\mathrm{W}}$ WKL. Bounded depth and unbounded width present a single site's candidate set as a subset of $\mathbb{N}$, given negatively by the clause's vanishing; a μ-clause searches it, and choosing an element is $C_{\mathbb{N}}$. The incomparability is standard. □

Remark 4 (Why the order on scheduler types is a lattice)*.* [ChSch] insists that the Weihrauch lattice is a lattice and not a chain, and calls this "the declarative shadow of the jumble-not-tower ontology" of [Agcy]. Proposition 4 gives the shadow a syntactic source: two independent axes of a formula — how deep it recurses and how wide it looks — generate incomparable degrees. A programmer can make a scheduler stronger in two ways that do not compare, and the non-comparison is visible in the clause without computing anything.

6 The formulae

We now write a clause schema for each degree. Throughout, the surface form is the same:

```
for( ptrn <- chan where psi ) P
```

and only `psi` changes. We give, for each degree, the schema, the shape of term it sits in, and the obligation it places on the resolver.

6.1 The bottom: finite presented choice

$$\psi \ ::= \ \lceil\beta\rceil \mid v \mid \psi \otimes \psi \mid \psi \oplus \psi \mid \langle K\rangle_{\mathcal{V}}^{\leq n}\psi$$

— no fixed point, or a budgeted one; finite candidate width. By Proposition 2 the family is presented as a list, and by Proposition 1(i) a crisp clause over an idempotent $\oplus$ is realised by a definable selection function. This is the Martin-Löf point of [ChSch]: the proof *is* the choice function, and no scheduler primitive is needed. Every clause in [GWhere] is of this shape.

Resolver obligation: none. Any deterministic tie-break realises it.

6.2 $\mathrm{LLPO} \equiv_{\mathrm{W}} C_2$: the negatively-presented binary commit

The weakest nontrivial degree is choice on two points, and it is reachable with two messages and no fixed point at all — provided the crisp guards are *semi-decidable complements* rather than decidable predicates.

```
new a, b in
  ( search(f, a) | search(g, b)
  | for( x <- a ; y <- b  where  [halts(f)] (+) [halts(g)] ) P )
```

where `search` produces on its channel exactly if its argument's search terminates, and the clause's two crisp conjuncts are the corresponding Σ^0_1 statements. The promise "not both" is the term's invariant; the clause's support is therefore never empty, and a resolver must commit to one side without being able to decide which side is safe.

We checked both halves computationally (`choicesim.py`, E3). Over $3 \times 40 \times 60$ instance/stage pairs the support was empty 0 times: the family is genuinely a family of nonempty sets. And for every stage-k selector, $k = 0, \ldots, 7$, under both tie-breaking preferences, the diagonal instance — which produces at stage $k+1$ on exactly the side the selector commits to — defeats it. So the clause is inhabited but has no stage-uniform computable section.

Resolver obligation: LLPO. This is the degree of "choose which of two Σ^0_1 statements fails".

Remark 5 (Rho programmers write this by accident). The term above is an ordinary race between two producers whose production is conditional on a semi-decidable event. Nothing exotic is involved and nothing in the current language flags it. That a routine racing idiom sits strictly above the computable degrees is the clearest argument for wanting the type in the first place.

6.3 $C_{\mathbb{N}}$: the width engine

$$\psi \ = \ \mu X.\left(\lceil\beta\rceil \oplus \langle K\rangle_{\mathcal{V}} X\right)$$

at a site whose channel accumulates unboundedly many data. The μ is an unbounded search through the candidate set for one satisfying β, where β is again only semi-decidable, so the set is given negatively: a candidate is excluded when it is *seen* to fail, and never positively certified in finite time.

```
new c in
  ( enumerate(c)                                  // unboundedly many data on c
  | for( x <- c  where  mu X. ([not_excluded(x)] (+) <K>_V X) ) P )
```

Resolver obligation: $C_{\mathbb{N}}$ — countable choice, an unbounded search that terminates because the family is promised nonempty, with no bound computable in advance on how long it takes.

6.4 $\mathrm{WKL} \equiv_{\mathrm{W}} C_{2^{\mathbb{N}}}$: the depth engine

$$\psi \;=\; \nu X.\, \big(\lceil\beta\rceil \otimes \langle K\rangle_{\mathcal{V}} X\big)$$

at a site of bounded width — two candidates suffice — under a recursive process, so that the candidate tree is infinite and finitely branching. The ν asserts the existence of an infinite run along which β never fails: exactly an infinite branch of an infinite binary tree.

```
new c in
  ( bits(c)                                       // two data at a time, forever
  | for( x <- c  where  nu X. ([ok(x)] (x) <K>_V X) ) P )
```

Resolver obligation: WKL — compactness. The resolver must produce a branch of a tree all of whose finite paths extend, which is not computable from the tree.

Remark 6 (The polarity of the fixed point is the polarity of the principle). μ searches and ν survives; $C_{\mathbb{N}}$ searches an unbounded set and WKL selects an infinite path. The syntactic distinction between the two fixed points and the semantic distinction between the two incomparable choice principles are the same distinction, and Proposition 4 is that observation made into a claim.

6.5 WWKL: the $\mathbb{R}_{\geq 0}$ row, and where the two axes touch

Take the WKL schema and replace the crisp β by a genuinely graded value, with $\mathcal{V} = \mathbb{R}_{\geq 0}$ and the clause normalised so that the total over a level is 1:

$$\psi \;=\; \nu X.\, \big(w(\cdot) \otimes \langle K\rangle_{\mathcal{V}} X\big), \qquad w : \mathrm{Cand} \to \mathbb{R}_{\geq 0}.$$

The set of surviving branches is now not merely nonempty but of *positive measure*, and the resolver's obligation weakens accordingly: it need only find a branch in a positive-measure closed set. That is WWKL, positive-measure choice, which Brattka, Gherardi and Hölzl identify as the degree of Las Vegas computability [BGH] — randomised computation that never errs but may not terminate.

This is the one place the two gradings of Principle 2 genuinely touch. The lift of §3.3 — selecting into distributions over candidates rather than into candidates, which is forced by Proposition 1(iii) — is precisely the randomness that WWKL measures.

Proposition 5 (The stochastic row is the randomised degree)**.** *A clause over $\mathbb{R}_{\geq 0}$ with a ν of unbounded depth, normalised and with positive total mass at every level, has choice type* WWKL, *and the resolver realising it is the distributional lift of §3.3: sampling.*

Read backwards, this says something about the Gillespie regime of [WGSLT]: a stochastic simulator is not a weaker classical scheduler but a scheduler at a specific, nonzero altitude, and the altitude is the price of the random bits.

6.6 $\lim$: plasticity without a modulus

The forager of [GWhere] maintains a belief whose value moves as evidence accumulates: the clause is evaluated against the current term, so its value at a site depends on what earlier firings did. Suppose the sequence of values converges — the belief settles — but no modulus of convergence is computable from the term. Then a resolver that must act on the *limit* value is computing a limit of a computable sequence, and $\lim$ is Weihrauch-equivalent to the Turing jump [BGP].

$$\psi \;=\; \nu X.\,\big(\mathrm{pow}(s,c) \otimes \langle K\rangle_{\mathcal{V}} X\big)$$

with $\mathrm{pow}(s,c) = s \cdot c$ the PLN strength-times-confidence value of [GWhere], $c = n/(n+k)$, and n the evidence count accumulated by firing.

Resolver obligation: $\lim$, if the resolver must act on the limit; $\perp$ if it acts on the current approximation. The distinction is the whole content: *learning is free, but knowing that you have finished learning is the jump.*

Remark 7 (The personality parameter, revisited). [GWhere] finds the PLN scepticism parameter k has an interior metabolic optimum. In the present reading, k is a *modulus policy*: it trades the rate at which confidence accumulates against the risk of acting on an unsettled value. That an optimum exists is the statement that neither $\perp$ nor $\lim$ is the right altitude to run at, and that the environment prices the compromise.

6.7 DC: the ω-fold product

Dependent choice requires no new syntax whatever. Per-match evaluation against the current term already makes a site's clause depend on the outcomes of earlier sites, which is exactly the defining feature of DC: later fibres depend on earlier selections.

Proposition 6 (Plasticity is dependent choice)**.** *A graded clause evaluated per match against the running term, at a sequence of sites chained through $\langle K\rangle_{\mathcal{V}}$, presents the family $\prod_{i\in\omega} X_i$ with X_{i+1} depending on the selection made at X_i. Its resolver is the ω-fold Escardó–Oliva product of the per-site resolvers, i.e. bar recursion.*

This is Conjecture 1 instantiated at ω, and it is the cleanest alignment in the note: [ChSch]'s row "countable/dependent choice DC — bar recursion; scheduler with memory and lookahead" is satisfied by a feature [GWhere] already has and describes as a convenience ("plasticity is free").

6.8 $C_{\mathbb{N}^{\mathbb{N}}}$: both engines together

Unbounded depth *and* unbounded width — a ν over a tree with infinite branching — presents the family as a closed subset of Baire space, and $C_{\mathbb{N}^{\mathbb{N}}}$ is the strongest of the closed-choice family.

$$\psi \;=\; \nu X.\,\big(\lceil\beta\rceil \otimes \mu Y.(\langle K\rangle_{\mathcal{V}}(X \oplus Y))\big)$$

Degree	Clause shape	Depth	Width	Resolver must
$\perp$ (definable)	no fixed point	$< \omega$	$< \omega$	tie-break
$\perp$ (optimising)	graded, idempotent $\oplus$	$< \omega$	$< \omega$	optimise
LLPO $\equiv_{\mathrm{W}} C_2$	crisp, Σ^0_1 guards	$< \omega$	2	commit blind
$C_{\mathbb{N}}$	$\mu X.(\lceil\beta\rceil \oplus \langle K\rangle_{\mathcal{V}} X)$	$< \omega$	ω	search unboundedly
WKL $\equiv_{\mathrm{W}} C_{2^{\mathbb{N}}}$	$\nu X.(\lceil\beta\rceil \otimes \langle K\rangle_{\mathcal{V}} X)$	ω	$< \omega$	be compact
WWKL	as WKL, $\mathcal{V} = \mathbb{R}_{\geq 0}$, normalised	ω	$< \omega$	randomise
lim	plastic, acting on the limit	ω	$< \omega$	take a limit
DC	chained plastic sites	ω	any	bar-recurse
$C_{\mathbb{N}^{\mathbb{N}}}$	ν over μ	ω	ω	choose on Baire space
AC	aperture, or $\langle @\rangle_{\mathcal{V}}$ under ν	—	—	be a choice function

Table 1: One `where` slot at every altitude. Only the formula changes.

— the inner μ searching an unbounded width at each level, the outer ν surviving unbounded depth.

Resolver obligation: $C_{\mathbb{N}^{\mathbb{N}}}$. This is the ceiling for closed terms, as §7.1 makes precise.

6.9 The master table

7 Reaching full AC

7.1 The ceiling for closed terms

First the obstruction, stated so that the routes past it can be seen to be the only ones.

Proposition 7 (Ceiling). *Let P be a closed term all of whose channels are fed by sub-terms of P, and let ψ be a graded clause containing no $\langle @\rangle_{\mathcal{V}}$ under a fixed point. Then the union of the candidate sets arising at any site over any run of P is recursively enumerable from P, the family is presented, and $\chi \leq_{\mathrm{W}} C_{\mathbb{N}^{\mathbb{N}}}$.*

Proof sketch. Candidates are assignments of datum occurrences to patterns. Data arise only by reduction from P, and the reduction relation of the rho calculus is r.e.; names are quoted processes, so the namespace generated by P is r.e. as well. Hence the total candidate family is an r.e.-indexed family of subsets of an r.e. set, given negatively by the clause's vanishing — a closed subset of a computably admissible, countably based space. Closed choice on such a space reduces to $C_{\mathbb{N}^{\mathbb{N}}}$. □

So a closed rho term, however baroque, cannot demand more than choice on Baire space. The set-theoretic AC — "survey an unindexed contention and select coherently" [ChSch] — needs a family that is not presented at all. There are exactly two ways for a clause to index one.

7.2 Route E: the aperture

The hypothesis of Proposition 7 that does most of the work is "all of whose channels are fed by sub-terms of P". Drop it.

Definition 5 (Aperture channel). A channel c of P is an *aperture channel* when the data on c are supplied by the environment — in the context-labelled semantics of [Probe], by the context K in $K[P]$ rather than by P — and no presentation of the possible supplies is given.

Construction 1 (AC at an aperture). `for( x <- sensor  where  psi(x) ) P`
where `sensor` is an aperture channel. Cand is indexed by whatever the environment supplies; the clause's support carves out a subfamily of it; and a resolver must select from a family over an index set with no presentation. That is full AC.

This is not a trick. It is [Agcy]'s own thesis rendered syntactically. That essay's central structural claim is that a confluent interior carries no choice resource, that "the resource degree of a macro-component is carried entirely by its boundary", and that "capacity, like choice, lives at the apertures". Route E says the same thing about clauses: *the only place in a program where full* AC *can be demanded is a boundary at which the world, not the term, supplies the contention.* A shard's oracle-consuming sites are exactly its I/O sites, and they are syntactically identifiable.

Remark 8 (This is also a security statement). If full AC is reachable only at apertures, then a static analysis that identifies aperture channels bounds the choice type of an entire shard. The unbounded altitude of a deployment is confined to its declared boundary, which is precisely the discipline a capability system already wants to impose for other reasons.

7.3 Route R: reflection, and Krivine's `quote`

The second hypothesis of Proposition 7 is that ψ contains no $\langle @ \rangle_{\mathcal{V}}$ under a fixed point. Drop that instead, and the family becomes impredicative.

Construction 2 (AC by reflection).

$$\psi \;=\; \nu X.\, \big(\langle @ \rangle_{\mathcal{V}} X \otimes \lceil \beta(*y) \rceil\big)$$

at a site binding y. The graded namespace modality evaluates X at the process $*y$ quoted by the received name. Since names are quotations of processes and processes may carry clauses, the domain over which $\langle @ \rangle_{\mathcal{V}}$ ranges includes processes whose own clauses are instances of ψ. The fixed point is therefore taken over a domain whose description mentions the fixed point being taken.

```
for( y <- c  where  nu X. ( <@>_V X (x) [beta(*y)] ) ) P
```

The family here is not presented in advance of the run, and not presented *by* the run either: it is specified by a condition quantifying over a domain that contains the condition. This is the shape of full AC as [ChSch] describes it — an unindexed contention surveyed coherently — rather than of any C_X for a fixed represented X.

Conjecture 2 (Degree of the reflective clause). *Construction 2 is not Weihrauch-reducible to $C_{\mathbb{N}^{\mathbb{N}}}$; its resolver requires a choice function on an arbitrary family, i.e. full* AC.

We flag this as a conjecture rather than a proposition because making it precise requires a representation of the impredicative family, and the natural candidates either collapse it back to $C_{\mathbb{N}^{\mathbb{N}}}$ by stratifying the namespace, or fail to be representations at all. The stratification question is the same one that returns in §10, which we take as evidence that it is the real content rather than a technicality.

There is, however, a striking piece of external corroboration. Krivine's classical realizability interprets dependent choice using a `quote` instruction — a primitive that assigns a code to a term, breaking the purely functional discipline of the abstract machine [Kri]. [ChSch] already notes this as "the cleanest existing statement that a choice-type's inhabitant is a program with matching extra power". The observation to add here is that *the rho calculus has* `quote` *as a term former.* What Krivine must add to his machine to realise choice, $@P$ already is.

Reflection is not an incidental convenience of the rho calculus that happens also to enable Construction 2. It is, on this reading, the calculus's native choice primitive, and it has been sitting in the syntax since [Mer05].

7.4 Exactly two routes

Corollary 2 (Trichotomy)**.** *Let P be a term and ψ a clause at a site of P. Exactly one of the following: (i) P is closed with presented channels and ψ has no $\langle @ \rangle_{\mathcal{V}}$ under a fixed point, and then $\chi \leq_{\mathrm{W}} C_{\mathbb{N}^{\mathbb{N}}}$ by Proposition 7; (ii) the site's channel is an aperture, and χ may be* AC *by Construction 1; (iii) ψ is reflective, and χ may be* AC *by Construction 2.*

The corollary is what makes the ceiling useful rather than discouraging. It is not that AC is unreachable; it is that the two ways of reaching it are both syntactically conspicuous. One is an I/O boundary and the other is a clause that quotes. A checker that flags those two features flags every site in a program that can demand a genuine choice function.

7.5 The Diaconescu cost

Diaconescu's theorem says that in a topos the axiom of choice implies the law of excluded middle. The analogue here is a cost, and it is worth stating because it is the price of the answer to the question this section asks.

Principle 3 (AC and interference do not cohabit)**.** Suppose a resolver realises full AC over the graded families a language of clauses defines: every such family admits a section produced by that resolver. Then the resolver's outcome at every site is a single candidate — a point — so the coefficient it assigns to any outcome is the value of the clause at one candidate, and no cancellation between candidates can occur. Hence the resolver exhibits no interference deficit, and it is insensitive to any structure in $\mathcal{V}$ beyond the support of the clause. An AC-realising resolver reads its clauses crisply.

Sketch, and its limits. A choice function selects an element of the fibre. The interference phenomenon of §8 is the assignment of coefficient 0 to an outcome reachable along two candidates with nonzero clause values; that requires the resolver to sum across candidates rather than select one. The two are exclusive by definition of "section". What is *not* established here is the converse Diaconescu direction — that the availability of choice forces the crisp sort's excluded middle as a theorem of the logic — which needs the subobject structure of a topos of graded predicates that we have not built. We state the forward direction only. $\square$

Remark 9 (Why this is a feature)*.* The two research programmes this note joins want different things: [ChSch] wants the strongest choice principles typed and bounded, [GWhere] wants interference arranged deliberately. Principle 3 says these are demands on different resolvers, not competing demands on one. A shard may run an AC-strength resolver at its apertures and a coherent resolver in its interior; what it may not do is expect one resolver to be both. The next section says why in the lattice's own terms.

8 Interference is transverse to the lattice

The complex row of the algebra table is not a high altitude. It is not an altitude at all.

Definition 6 (Interference deficit [GWhere])**.** The *interference deficit* of a term P is $\mathrm{Reach}(P) \setminus \mathrm{supp}([\![P]\!])$: the outcomes operationally reachable from P that carry coefficient 0 in the denotation.

Proposition 8 (A deficit is not a section)**.** *If the interference deficit of P at a site is non-empty, no selection function on* Cand *produces the resolver's behaviour at that site.*

Proof. A section of a family lands in a fibre: whatever a choice function returns is an element of the set it was asked to choose from, and the outcome that element produces therefore carries positive weight. An outcome in the deficit is reachable — some candidate produces it — but carries coefficient 0. So the resolver's behaviour is not the image of any section. □

The mechanism is the one already worked out in [GWhere]: two candidates whose outcomes coincide contribute $w_0 \oplus w_1$ to that outcome, and over $\mathbb{C}$ the sum of two nonzero terms can vanish. We recomputed the minimal case (`choicesim.py`, E5): with $|w_0| = |w_1| = 1/\sqrt{2}$ and relative phase 0, π, $\pi/2$ the coefficient is 1.4142, 0.0000, $0.7071{+}0.7071i$, giving $|\cdot|^2$ of 2.000000, 0.000000, 1.000000. Over $\mathbb{R}_{\geq 0}$ this cannot happen: minimising $w_0 + w_1$ over 100×100 strictly positive pairs gives 0.020000, the smallest pair, and never 0.

Principle 4 (Two axes, one crossing)**.** The Weihrauch axis measures *how hard it is to produce a witness*. The value-algebra axis measures *whether "which candidate?" has an answer*. Over $\mathbb{B}$, an idempotent graded algebra, or $\mathbb{R}_{\geq 0}$, it does, and the resolver is a section, possibly randomised; the choice hierarchy applies and the schemas of §6 classify it. Over $\mathbb{C}$ it does not: the resolver sums rather than selects, and destructive interference *deletes* a reachable witness, which no section can do. The two axes cross at exactly one point, WWKL, where the lift into distributions is still a lift into sections-with-randomness.

Remark 10 (Consistency with Principle 3)*.* The two statements are the same fact from opposite ends. AC forces a section, which forbids a deficit; a deficit forbids a section, which forbids AC. The top of the choice hierarchy and the interfering regime are disjoint, and neither is above the other.

9 Fairness disciplines are clause combinators

[ChSch] places fairness on the typing side: a fairness discipline constrains which schedulers inhabit a choice-type. In the graded setting it moves to the term side, because a clause can starve a candidate all by itself.

Definition 7 (Support fairness)**.** A run is *weakly support-fair* when a candidate continuously in $\mathrm{supp}([\![\psi]\!])$ eventually fires, and *strongly support-fair* when a candidate infinitely often in the support eventually fires.

Proposition 9 (Zero is absorbing)**.** *A receipt whose clause evaluates to* 0 *is outside the support, hence never fires, hence never changes the term against which its clause is evaluated. If the clause's value depends only on state that firing would update, the value remains* 0 *forever.*

This is not hypothetical. [GWhere]'s forager exhibits it: with the PLN confidence $c = n/(n+k)$ and no prior evidence, $n_0 = 0$ gives $c = 0$, gives clause value 0, gives a receipt that never fires and therefore never acquires the evidence that would raise n. We reproduced the trap in a stripped-down model (`choicesim.py`, E4): with $n_0 = 0$ and no repair, 0 firings over the horizon and a terminal stack of 0.00; with $n_0 = 1$, 197 firings and 89.20.

The repair [GWhere] found is a disjunct, and in the present language that is literally a formula.

Definition 8 (The exploration combinator)**.** For $\epsilon \in V$ with $\epsilon \neq 0$, write $\psi^\epsilon = \psi \oplus \epsilon$.

Proposition 10 (Weak fairness is a combinator)**.** *Over an algebra with no additive inverses,* $\mathrm{supp}(\psi^\epsilon) = \mathrm{Cand}$, *so every enabled candidate is in the support at every step and every run is weakly support-fair.*

Measured: the same forager with $n_0 = 0$ and $\psi \oplus 0.02$ recovers to 179 firings and a terminal stack of 83.80, against 0 and 0.00 without it (`choicesim.py`, E4). Exploration is a disjunct; fairness is a formula.

Remark 11 (Strong fairness is a fixed-point condition, and it costs). Weak support fairness is cheap because it is a condition on each step. Strong support fairness is a condition on infinite runs — "infinitely often in the support implies eventually fires" — which is a ν over the run tree, and by §6.4 a ν over an infinite finitely-branching tree is WKL. So the classical folklore that strong fairness is a substantially stronger assumption than weak fairness acquires a degree: the gap is compactness.

Remark 12 (The additive-inverses caveat is the interference caveat). Proposition 10 requires that adding ϵ cannot cancel. Over $\mathbb{C}$ it can: $\psi \oplus \epsilon$ may vanish where ψ did not. So the exploration combinator is a fairness device in the $\mathbb{B}$, $\mathbb{R}_{\geq 0}$ and PLN regimes and a *de-tuning* device in the coherent one. The same structure that buys interference costs the cheap fairness repair.

10 Subject reduction, and reflection as the suspect

[ChSch]'s central metatheorem is subject reduction, read as: *scheduling cannot bootstrap an oracle*. [Agcy] makes the same question the hinge of its factorisation — if $\chi(c)$ can rise as a term runs, an interior cut can cross the threshold and an island cracks. In the present setting the question becomes concrete and, we suspect, has a negative answer for the full language.

Definition 9 (Clause level)**.** The *level* $\ell(\psi)$ of a clause is its position in Table 1, read syntactically: the polarity and boundedness of its fixed points, the width of the sites it ranges over, and whether it contains $\langle @ \rangle_{\mathcal{V}}$ under a fixed point.

Question 1 (Subject reduction)**.** *If $P \to P'$ by a firing at a site whose clause has level ℓ, is every clause in P' of level $\leq \ell$?*

Proposition 11 (Reflection defeats it)**.** *No. Let Q be a process containing a receipt whose clause is* $\nu X.(\lceil \beta \rceil \otimes \langle K \rangle_{\mathcal{V}} X)$, *at level* WKL. *Let*

$$P \;=\; c!(Q) \;\mid\; \mathrm{for}(y \leftarrow c \,\mathrm{where}\, \lceil \mathrm{true} \rceil) * y.$$

Every clause in P *is fixed-point-free, so* $\ell(P) = \bot$. *But* $P \to *@Q \equiv Q$, *and* $\ell(Q) = \mathrm{WKL}$. *The level rose across a reduction whose own clause was trivial.*

Proof. Immediate from $*@Q \equiv Q$ and the definition of level as a syntactic property of the clauses present. □

The counterexample is embarrassingly cheap, and that is the point: the mechanism is not some subtle cascade but the basic move of the calculus. A name is a quoted process; a process may carry clauses; dropping a received name installs them. *Reflection lets a low-altitude term receive a high-altitude one.* In [Agcy]'s language, the island cracks not because a boundary decision manufactured strength inside, but because strength was mailed in.

Remark 13 (The same suspect, three times). Reflection is now the suspect in three separate open questions of this programme. It is the conjectured source of length-asymmetric reconvergence, without which interference in unmodified rho is purely constructive [GWhere]; it is Route R to full AC (§7.3); and it is the counterexample to subject reduction here. The recurrence is not a coincidence: all three are consequences of the namespace being able to carry arbitrary computational content, which is what $@P$ is for. Krivine's `quote` is the same primitive and it too is what buys choice and what breaks the standard metatheory.

10.1 The repair, and where AC ends up

Definition 10 (Stratified `where`)**.** A shard declares a level ceiling L. A clause *elaborates at level* L when (i) its own shape is at level $\leq L$ by Definition 9, and (ii) every occurrence of $*y$ in it, for y bound by the site, is guarded by a level check on the received process — refusing at run time any received term carrying a clause above L.

Proposition 12 (Subject reduction, stratified)**.** *In the fragment where every clause elaborates at level L and every drop is level-checked, Question 1 has a positive answer, and the choice type of a term is a reduction invariant bounded by L.*

Proof sketch. The only way to introduce syntax not present in P is substitution of a received name followed by a drop; (ii) refuses drops that would raise the level; (i) bounds what is present initially; and no other rule of the calculus creates clauses. □

Corollary 3 (AC is outside)**.** *Route R (Construction 2) is by definition unstratified, and Route E (Construction 1) violates the hypothesis of the ceiling proposition rather than the stratification, so it survives stratification but only at declared aperture channels. Hence in a stratified shard, full* AC *occurs exactly at declared apertures, and the choice type of the interior is invariant.*

This is the answer we would want if we could choose it. The factorisation of [Agcy] is stable in the stratified fragment; AC is available, and available precisely where that essay says capacity lives; and the failure mode is not hidden but is a declared feature of a channel. Whether the stratified fragment is expressive enough to be worth programming in is the obvious next question and we do not answer it.

11 Implementation

What changes, concretely, in rholang 1.4 and MeTTaIL [Omni].

I1. **The graded sort grows fixed points.** Definition 3: add X, $\mu X.\psi$, $\nu X.\psi$ and $\langle @ \rangle_{\mathcal{V}}\psi$ to the graded condition grammar, with guardedness checked at elaboration. The crisp sort is unchanged.

I2. **Unfolding is budgeted by default.** Remark 3: the existing inference-token budget caps unfoldings, which keeps every clause at $\perp$ unless the programmer explicitly buys altitude. This makes the safe default the cheap default.

I3. **The resolver is an interface, not a fixed engine.** A resolver is a J-algebra: given a clause's value vector, return a candidate (or a distribution over candidates). The existing RSpace race, the Gillespie sampler of [WGSLT], and a Viterbi or tropical optimiser are four instances. Attainment (Definition 2) is the conformance law such an interface should carry, in the style of the existing `oslf.rs` conformance-law idiom.

I4. **Level as an elaboration-time judgement.** Definition 9 is computed from the clause's syntax tree; Definition 10 adds the drop check. Both are decidable and neither requires running anything. This is the same shape as the isometry judgement proposed in [GWhere] for the coherent fragment, and the two should share machinery.

I5. **Aperture channels are declared.** Route E needs a channel attribute distinguishing environment-fed channels from term-fed ones, which a capability system plausibly wants anyway. Corollary 3 then makes the shard's maximal choice type a static read-off.

I6. **Candidate identity must be by occurrence.** As [GWhere] already notes, an implementation keying data by content, or allocating identifiers during firing, cannot represent Cand faithfully — and the width engine of §5 is precisely a statement about $|\mathrm{Cand}|$, so the addressing defect would corrupt the level judgement as well as the values.

12 Open problems

O1. **Degrees, properly.** Fix a representation of the candidate family as a point of a represented space and prove that each schema of §6 has the degree claimed, rather than merely imposing the corresponding obligation.

O2. **Conjecture 1.** Prove that the Escardó–Oliva product, the Weihrauch composition $\star$, and chaining through $\langle K\rangle_{\mathcal{V}}$ agree, on the attainable fragment. This is what makes DC the ω-fold product and bar recursion the scheduler.

O3. **Conjecture 2.** Settle the degree of the reflective clause. Either exhibit a representation collapsing it to $C_{\mathbb{N}^{\mathbb{N}}}$ — in which case Route R is not a route and AC is reachable only at apertures — or show the impredicativity is essential.

O4. **Diaconescu, properly.** Build the topos of graded predicates over a GSLT in which Principle 3 is a theorem with its converse, rather than a forward implication and an analogy.

O5. **Expressiveness of the stratified fragment.** Proposition 12 buys subject reduction by refusing drops that raise the level. Is the resulting language still expressive enough for the applications — in particular, does it retain the self-hosting and DSL-mobility properties that motivate reflection in the first place?

O6. **Non-attainable resolvers as a monad.** §3.3 lifts the selection function's codomain twice, to distributions and to amplitudes. Is there a uniform account — a parameterised selection monad over a choice of enrichment — of which the classical, randomised and coherent resolvers are the three instances?

O7. **The altitude of consensus.** [ChSch] proposes stacking its pipeline with the Coalition-Logic consensus extraction. With the present schemas one can ask a sharper question: what is the level, in the sense of Definition 9, of the clauses a correct-by-construction consensus protocol requires, and is it below the ceiling a shard would want to declare?

13 Conclusion

The scheduler calculus that [ChSch] lists as an open problem is, we propose, the graded `where` clause of [GWhere] with its fixed point restored. A clause is a quantifier and a resolver is a selection function; the monad morphism between them says when the resolver realises the clause; and attainability of the induced quantifier is a sharp line, computable, and drawn exactly where the algebra's addition stops being idempotent. Below that line the scheduler is a pure term. Above it the scheduler must optimise, or randomise, or — in the coherent regime — stop being a scheduler at all.

The ascent through the choice hierarchy is then the reintroduction of a single syntactic feature, the graded fixed point, driven by three engines: depth, width, and reflection. Depth and width are incomparable, which is why the order on scheduler types is a lattice, and the incomparability is visible in the formula rather than in the metatheory. Full AC is reachable, by exactly two routes, and both are conspicuous: a channel fed by the world, or a clause that quotes. The first is where [Agcy] already said capacity lives. The second is what Krivine had to add to his machine and what the rho calculus has had since 2005.

Two facts sit at the edges of the picture and both are, we think, load-bearing rather than awkward. The complex-valued regime is transverse to the hierarchy, not high in it, because a resolver that interferes deletes a reachable witness and no section can do that. And the mechanism that buys full AC is the mechanism that breaks subject reduction, which is [Agcy]'s open hinge with a two-line counterexample and a stratification that repairs it. Taken together they say that a shard can have any two of unrestricted reflection, invariant choice types, and coherent resolution — and that the choice of which two is a language-design decision that can now be written down, checked at elaboration, and read off a clause.

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A The computations

All numbers in this note are produced by `choicesim.py`, which has no dependencies beyond the Python standard library. It runs five experiments.

E1, attainability. For each algebra and clause, tests over $20{,}000$ random payoffs whether $\phi_\psi(p)$ is a value of p; then verifies Proposition 1(ii) by checking that "attained" agrees with "the winning candidate carries the unit" (100.00%); then verifies the distributional lift by Monte Carlo over $400{,}000$ draws.

E2, the product. Computes the Escardó–Oliva product of two argmax selection functions and the chaining of two graded sites through $\langle K \rangle_\mathcal{V}$ over $20{,}000$ random payoff tables (0 disagreements), with a negative control that chains in the wrong algebra ($4{,}483$ disagreements) so the agreement is not vacuous.

E3, LLPO**.** Checks that the support of the two-sided race is never empty over $3 \times 40 \times 60$ instance/stage pairs, and constructs, for each stage-k selector with $k \leq 7$ and each tie-break, the diagonal instance defeating it.

E4, fairness. Runs the stripped-down forager in three configurations, exhibiting the bootstrap trap (0 firings, terminal 0.00) and its repair by $\psi \oplus \epsilon$ (179 firings, terminal 83.80) against the primed baseline (197, 89.20).

E5, the deficit. Computes the coefficient on the coinciding outcome at relative phases 0, π, $\pi/2$, and minimises the corresponding $\mathbb{R}_{\geq 0}$ sum over 100×100 strictly positive pairs to confirm it never vanishes.